EXPERIMENT 3
CENTRIFUGAL PUMP SRF MODEL

Computational Fluid Dynamics Laboratory Report

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1. Aim

To study the pressure rise, pump head and thermal behaviour of a simplified two-dimensional annular centrifugal-pump model using a single rotating frame (SRF), and to test the expected near-linear scaling of pump head with rotational speed squared.

2. Objectives

- Run the solver at 1000, 2000 and 3000 RPM while keeping Reynolds number fixed at 500.
- Compare the computed pressure rise and pump head.
- Plot pump head against RPM^2 and evaluate the near-linear trend.
- Examine the residual and outlet-temperature histories.
- Visualize the pressure field and identify limitations of the teaching model.

3. Numerical method

The supplied Python solver models flow in a two-dimensional annular passage using a prescribed single-rotating-frame velocity field. It computes pressure from a simplified radial expression and solves a steady thermal transport equation iteratively. The temperature update is relaxed, and convergence is monitored using the maximum absolute change between successive temperature fields. The solver writes a CSV history, a text summary, and a structured-grid VTK field file.

4. Governing equations

Angular speed: $\omega = 2\pi N/60$, where N is rotational speed in RPM.

Inlet velocity from the model Reynolds number: $U_{in} = Re\ \mu/(\rho\ D_h)$, where D_h is the model hydraulic diameter.

Radial velocity prescribed by continuity-like scaling: $u_r = U_{in}\ r_i/r$.

Absolute tangential velocity: $u_{\theta,abs} = s\ \omega\ r$; relative tangential velocity: $u_{\theta,rel} = (s - 1)\omega r$, where s is the swirl parameter.

The supplied pressure model is: $p(r) = \frac{1}{2}\rho\,\omega^2(r^2 - r_1^2) - [12\,\mu\,U_in\,r_1/d^2]\,\ln(r/r_1)$.

Pump head: $H = \Delta p/(\rho g)$, with Δp the mean outlet pressure minus mean inlet pressure.

The thermal equation is a discretized advection–diffusion balance with wall heat exchange and a modeled loss-heating term. Its iterative update combines radial and angular advection, diffusion, wall heat transfer, and loss heating. This is the implemented teaching-model equation rather than a full Navier–Stokes energy solution.

5. Geometry, domain and boundary conditions

- Annular inner radius: $r_1 = 0.030$ m.
- Annular outer radius: $r_2 = 0.120$ m.
- Channel depth: $d = 0.010$ m.
- Radial coordinate: $r \in [0.030, 0.120]$ m; angular coordinate $\theta \in [0, 2\pi)$.
- Inner radial boundary: temperature fixed at $T_in = 300$ K.
- Outer radial boundary: zero-gradient temperature condition, implemented as T[-1,:] = T[-2,:].
- Angular direction: periodic treatment using array rolls.
- Pressure and velocity are prescribed analytically by the model; the simulation does not solve full pressure–velocity coupling.

6. Complete parameter set and grid

Parameter	Value	Notes
Radial grid, nr	100	Lab preset
Angular grid, nθ	160	Lab preset
Reynolds number	500	Fixed in all runs
Rotational speeds	1000, 2000, 3000 RPM	Compulsory comparison
Inlet temperature	300 K	Fixed
Wall temperature	330 K	Fixed
Iterations	1200 maximum	All runs
Swirl parameter	0.85	Fixed
Wall HTC	120	Model input
Loss fraction	0.04	Fixed
Density, ρ	997 kg/m³	Hard-coded model property
Dynamic viscosity, μ	0.0010 Pa·s	Hard-coded model property
Specific heat, cp	4180 J/(kg·K)	Hard-coded model property
Thermal conductivity	0.60 W/(m·K)	Hard-coded model property
Gravity, g	9.81 m/s²	Head conversion

7. Compulsory assignment: 1000, 2000 and 3000 RPM

The three cases were run with the same grid (100 × 160), Re = 500, inlet/wall temperatures, swirl parameter, wall HTC, loss fraction and 1200-iteration limit. Only rotational speed was changed.

Speed (RPM)	RPM²	Pressure rise (Pa)	Pump head (Pa)	Final residual
1000	1,000,000	73,799.842	7.545556	3.560191e-04
2000	4,000,000	166049.8504	16.9775228	1.014674e-03
3000	9,000,000	664,199.577	67.910109	1.216679e-04

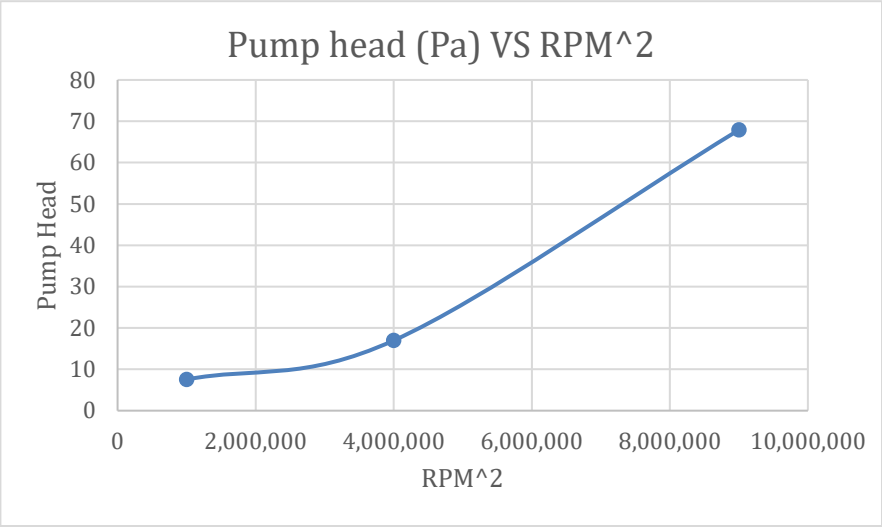


Figure 1. Computed pump head versus RPM² at fixed Re = 500.

8. Inline field visualization

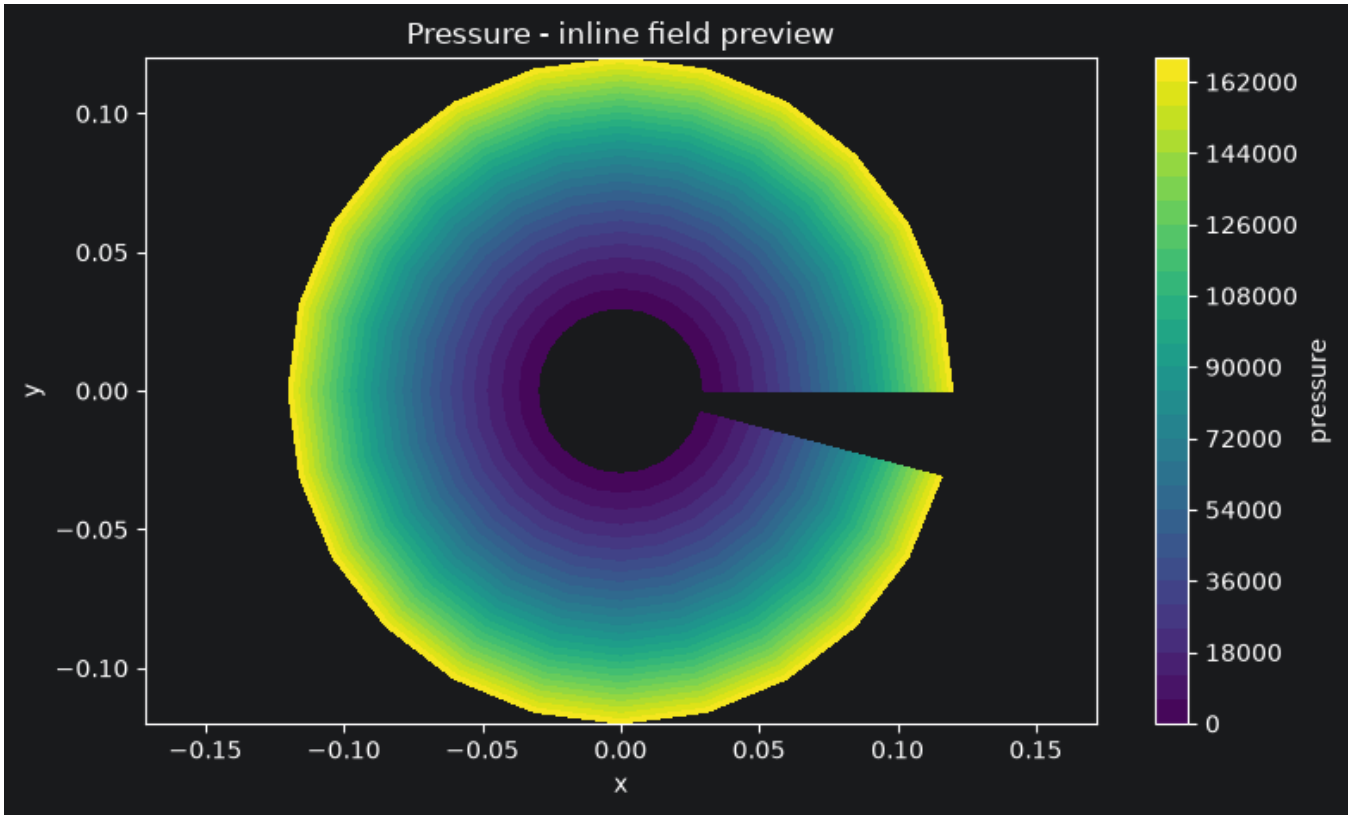


Figure 4. Pressure field from the 2000 RPM run (Re = 500); axes are x and y in metres and the colour bar is pressure in pascals.

The field is nearly axisymmetric because pressure is prescribed as a function of radius. Pressure increases from the inner radius toward the outer radius, consistent with the $\omega^2(r^2 - r_i^2)$ term in the model. This visualization reflects the analytical pressure prescription rather than a pressure field obtained from a fully coupled CFD solution.

9. Observation:Numerical Behaviour Versus Physical InterpretationThe pump head increased with rotational speed. The calculated values were 7.5456 m at 1000 RPM, 16.9775 m at 2000 RPM, and 67.9101 m at 3000 RPM. The pressure field showed an approximately axisymmetric distribution, with pressure increasing from the inner radius to the outer radius. However, the results do not follow the expected (H \propto N^2) relationship consistently. The residual values also differed between cases, indicating that convergence should be checked individually**.**

10. Limitations

The model uses a simplified two-dimensional annular geometry.

Velocity and pressure are prescribed analytically rather than calculated using fully coupled Navier–Stokes equations.

Heat transfer and losses are represented using simplified equations.

Grid independence and experimental validation have not been performed.

The residuals indicate that convergence requires further investigation.

Real pump performance factors, such as efficiency, flow rate, and impeller geometry, are not fully represented.

11. Conclusion

The experiment investigated the effect of rotational speed on pump head, pressure distribution, and thermal behaviour at a constant Reynolds number of 500. Pump head increased with rotational speed, and the pressure field showed the expected radial trend. However, the results do not establish a consistent square-law relationship between pump head and RPM. Further verification and convergence testing are required. The model is useful for understanding basic centrifugal pump behaviour but cannot accurately predict real pump performance without further validation.s